MEMBRIX — PROJECT PLAN

Internship project | Client: Thendral Yoga Loft | Initial plan

1. Project overview

Membrix is a membership-management web application planned for Thendral Yoga Loft. It aims to make membership selection, payment confirmation, digital pass checks and class-entry tracking easier for members and staff.

2. Users and problem

Members: create an account, view membership options, pay for a plan and access a QR membership pass. **Staff:** scan a member's QR pass, check its validity and record class entry. The project addresses the need to reduce manual membership verification and make class usage easier to track.

3. Planned features

Member experience	Signup/login; view three membership plans; membership status; QR pass.
Payments	Razorpay test-mode checkout; verify payment status on the server before activating a membership.
Staff tools	Scan QR pass; verify membership status; record class entry.
Membership rules	Support pausing memberships and an 8-class allowance for the Basic plan.
Automation and AI	Scheduled daily job; AI-assisted re-engagement message draft with a safe fallback if AI is unavailable.

4. Planned technology

Next.js with TypeScript and Tailwind CSS for the web app; Supabase for authentication and data; Razorpay test mode for payment integration; GitHub for version control and automated checks; Vercel for deployment. Service availability and free-tier limits will be checked before use.

5. Data and security approach

Use fictional/sample data during development. Keep secret keys out of frontend code and GitHub. Verify payments and QR pass validity on the server. Do not allow automated tests to make real payments or send real messages. AI message generation must not prevent the core membership flow from working.

6. Initial delivery approach

Build and validate the member-facing flow first, then add payment and QR verification, staff entry tracking, membership rules, scheduled automation and the AI-assisted message draft. Test each feature locally and through GitHub Actions before releasing updates.

Note: This is the initial project plan. Features and implementation details may be refined during development based on testing and client requirements.